Linear Functions & Inequalities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line $6x + y = 1$?

2 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = 7x - 11$, what is $f(-9)$?

3 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line has slope -8 and passes through the point $(3, -9)$. What is the y-intercept of the line?

4 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line has slope -1 and passes through the point $(1, -3)$. What is the y-intercept of the line?

5 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The table shows that $f(-1) = -8$ and $f(1) = 2$. Assuming f is linear, what is $f(3)$?

6 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = -7x - 4$, what is $f(6)$?

7 [Foundational] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line is parallel to the line $y = -4x + 3$ and passes through the point $(-7, -5)$. What is its y-intercept?

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The table shows that $f(1) = 10$ and $f(3) = 20$. Assuming f is linear, what is $f(5)$?

x	f(x)
1	10
3	20
5	?

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9 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line $3x + y = 17$?

10 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The point $(4, 14)$ lies on the line $y = kx - 2$ in the xy-plane, where k is a constant. What is the value of k?

11 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = 7x + 6$, what is $f(3)$?

12 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line has slope -7 and passes through the point $(-2, 12)$. What is the y-intercept of the line?

13 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = x + 12$, what is $f(-8)$?

14 [Intermediate] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = -7x + 4$, what is $f(2)$?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line in the xy-plane that passes through the points $(-3, 5)$ and $(1, -3)$?

16 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the horizontal line defined by $y = 7$ in the xy-plane?

17 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The table shows that $f(-4) = 2$ and $f(-2) = -4$. Assuming f is linear, what is $f(0)$?

18 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

If $f(x) = -7x - 11$, what is $f(-3)$?

19 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line passes through the points $(-1, 15)$ and $(3, 7)$. If $f(x)$ represents this line, what is $f(-6)$?

20 [Challenging] • [SPR] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line passes through the points $(2, 4)$ and $(-4, 10)$. If $f(x)$ represents this line, what is $f(-5)$?

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Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which values of x satisfy $-2x + 3 > 7$?

a. $x > -2$

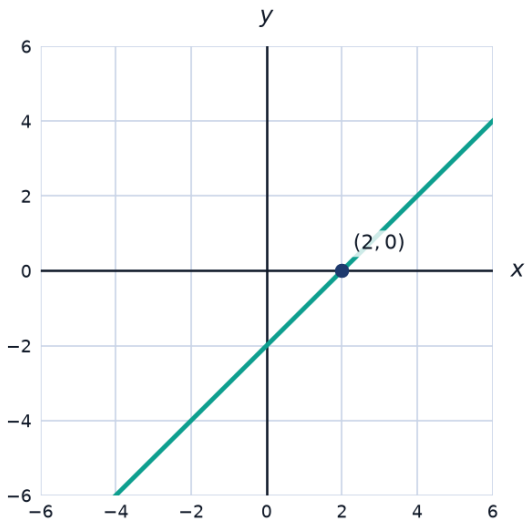
b. $x < -2$

c. $x > 2$

d. $x < 2$

22 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The graph of a function is shown. At which x-value does the function equal zero?



a. $x = -2$

b. $x = 0$

c. $x = 1$

d. $x = 2$

23 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line in the xy-plane has a slope of 3 and passes through the point $(2, 11)$. Which equation represents this line?

a. $y = 3x + 5$

b. $y = 3x + 11$

c. $y = 3x - 5$

d. $y = 2x + 3$

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24 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which of the following equations represents a horizontal line passing through the point $(3, -5)$?

a. $y = -5$

b. $x = 3$

c. $y = 3x - 5$

d. $y = -5x + 3$

25 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A gym charges a monthly membership fee of USD 30 plus USD 2 for each fitness class attended. If a member attends c fitness classes in a month, which expression represents the member's total cost for that month?

a. $30 + 2c$

b. $30c + 2$

c. $2(30 + c)$

d. $\frac{30 + c}{2}$

26 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A line has equation $y = -4x + 9$. What is the y-intercept of the line?

a. $(0, -4)$

b. $(0, 9)$

c. $(9, 0)$

d. $(-4, 0)$

27 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which inequality represents all values of x such that $4x - 5 \le 19$?

a. $x \le 4$

b. $x \le 5$

c. $x \le 6$

d. $x \le 7$

28 [Foundational] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line $2x + 5y = 20$?

a. $-\frac{2}{5}$

b. $\frac{2}{5}$

c. $-\frac{5}{2}$

d. 4

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Sarah earns USD 15/hour working h hours plus a USD 50 weekly bonus. Which expression is her weekly earnings?

a. $15 + 50h$

b. $15h + 50$

c. $50h + 15$

d. $65h$

30 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A taxi company charges an initial booking fee plus a fixed rate per mile driven. A trip of 4 miles costs USD 14, and a trip of 8 miles costs USD 22. What is the initial booking fee?

a. USD 4

b. USD 6

c. USD 8

d. USD 2

31 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which of the following lines is parallel to the line $y = 4x - 9$ in the xy-plane?

a. $y = 4x + 12$

b. $y = -4x - 9$

c. $y = -\frac{1}{4}x + 3$

d. $y = \frac{1}{4}x - 9$

32 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which value of x satisfies the inequality $-3x + 8 < -7$?

a. 3

b. 4

c. 5

d. 6

33 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A plumber's total charge in dollars for a job lasting h hours is given by $C(h) = 31h + 25$. What does the number 25 represent in this context?

a. the flat fee charged even if no labor hours are worked

b. the number of hours the plumber worked

c. the hourly rate the plumber charges

d. the total charge for a 1-hour job

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34 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A baker can produce at most 60 pastries each morning. If c represents the number of croissants and m represents the number of muffins, which inequality represents this restriction?

a. $c + m \leq 60$

b. $c + m \geq 60$

c. $cm \leq 60$

d. $c \leq 60m$

35 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which of the following points lies in the region defined by $y \geq x + 5$ in the xy-plane?

a. $(1, 8)$

b. $(2, 5)$

c. $(4, 7)$

d. $(3, 0)$

36 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which of the following inequalities is equivalent to $4x - 6y > 18$?

a. $y < \frac{2}{3}x - 3$

b. $y > \frac{2}{3}x - 3$

c. $y < -\frac{2}{3}x + 3$

d. $y > -\frac{2}{3}x - 3$

37 [Intermediate] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which point satisfies the inequality $y < x + 5$?

a. (-1, 4)

b. (-5, 5)

c. (5, 14)

d. (-5, -3)

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the y-intercept of the line $5x - 2y = 20$ in the xy-plane?

a. $(0, -10)$

b. $(0, 10)$

c. $(4, 0)$

d. $(-4, 0)$

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39 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

At what point does the line $y = -\frac{1}{2}x + 4$ intersect the x-axis in the xy-plane?

a. $(8, 0)$

b. $(-8, 0)$

c. $(0, 4)$

d. $(4, 0)$

40 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of any line in the xy-plane that is perpendicular to the line $y = -\frac{2}{5}x + 3$?

a. $\frac{5}{2}$

b. $-\frac{5}{2}$

c. $\frac{2}{5}$

d. $-\frac{2}{5}$

41 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

What is the slope of the line passing through $(2, 5)$ and $(6, 17)$?

a. 2

b. 3

c. 4

d. 5

42 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

The function $f(x) = 15x + 40$ models the total charge in dollars for x hours of equipment rental. What does the value 40 represent?

a. The hourly rate for the rental

b. The flat initial fee charged regardless of hours

c. The maximum number of rental hours

d. The total charge for 15 hours

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43 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A plumber's total charge in dollars for a job lasting h hours is given by $C(h) = 51h + 41$. What does the number 41 represent in this context?

a. the total charge for a 1-hour job

b. the flat fee charged even if no labor hours are worked

c. the number of hours the plumber worked

d. the hourly rate the plumber charges

44 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

Which point satisfies the inequality $y < 2x - 2$?

a. (-2, 0)

b. (-5, -7)

c. (-4, -10)

d. (-2, -9)

45 [Challenging] • [MCQ] • Target: Linear Functions, Slope-Intercept Modeling & Inequalities

A region in the xy-plane is bounded by a dashed line with y-intercept $(0, 3)$ and slope -2, and the region below this line is shaded. Which inequality represents this region?

a. $y < -2x + 3$

b. $y \leq -2x + 3$

c. $y > -2x + 3$

d. $y \geq -2x + 3$

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